Aditya Meenakshisundaram

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Seeking a Summer 2027 internship in battery and advanced-materials R&D, process development, or scale-up.

Education

Texas A&M University Expected May 2028
Bachelor of Science in Chemical Engineering
Cumulative GPA: 3.91 / 4.00 | Craig & Galen Brown Engineering Honors
Organizations: AIChE, Aggies in Tech, Omega Chi Epsilon

Work Experience & Research

Photolithography & Metrology Intern May 2026 – August 2026
Samsung Semiconductor Austin, TX

- Recovered 300–400 wafers/day of throughput using I-MR statistical process control to isolate a downstream track bottleneck caused by a degraded buffer-unit transfer mechanism
- Engineered a Python-based overlay-correction tool that transformed 500+ wafer- and shot-level metrology measurements per lot into scanner alignment corrections, saving 250 engineering hours annually
- Developed and deployed a PyTorch CNN on FFT-derived heatmaps to classify spin-coater defects with 84% accuracy, automating review of 1,000+ previously manual process images per day
- Automated lot-level overlay metrology skip-factor resets with Python and SQL, restoring process-specific defaults while saving 50 engineering hours per week

Samsung Semiconductor Research Fellow Jan 2026 – Present
Samsung Semiconductor & Artie McFerrin Department of Chemical Engineering College Station, TX

- Developed a passive thermal-switching composite for lithium-ion battery thermal-runaway mitigation, using indium phase change, $CaCO_3$ decomposition, and a retained titanium framework to transition from heat conduction to insulation
- Calculated thermal conductivity ($k = \alpha \rho C_p$) across 50–70 wt% indium formulations, identifying 65 wt% as the strongest-performing composition across thermal response and structural integrity
- Fabricated composites end to end through powder mixing, pellet pressing, and 900 °C heat treatment; analyzed cross-sectional morphology and conductive pathways by SEM across 4+ prototype iterations
- Built a verified 1D finite-volume model of a 5-cell battery stack in Python; swept 361 barrier designs and identified a simulation-predicted hot-state conductance threshold of $\sim 32\,\mathrm{W\,m^{-2}K^{-1}}$ for suppressing runaway propagation

Undergraduate Researcher, Green Lab Aug 2025 – Present
Artie McFerrin Department of Chemical Engineering, Texas A&M University College Station, TX

- Investigated converting liquid polycarbosilane (PCS) into electrically conductive SiC for RF susceptors, targeting the lowest pyrolysis temperature that still yields conductivity
- Processed PCS end to end through degassing, curing, resting, and pyrolysis; ran 10+ controlled campaigns at 800–1200 °C, varying temperature, ramp rate, and atmosphere
- Narrowed the onset of measurable electrical conductivity to 900–1000 °C via four-point probe, focusing subsequent trials on the identified transition range
- Achieved 92% SiC phase purity by XRD, examined backbone-fiber and PCS-coating cross-sections by SEM, and evaluated RF heating at 200–300 W to assess electromagnetic coupling

Projects

STEM Research Finder Co-Founder; Lead Developer
stemresearchfinder.tech

- Co-founded a research-discovery platform grown to 3,000+ users with no paid acquisition, indexing 7,864 STEM faculty across 8 universities, including Texas A&M, MIT, and Harvard
- Built Python scraping pipelines across 50+ university sources and LLM-powered resume matching that ranks faculty by fit and drafts personalized outreach emails

Substack Author Semiconductor & Technology Writer
adimeenak.substack.com

- Published 30+ technical articles on semiconductors, materials, and energy since Aug 2025, reaching 400+ monthly readers with no promotion budget
- Explained EUV lithography, advanced packaging and heterogeneous integration, and alternative semiconductor materials, connecting process physics and materials tradeoffs to manufacturing and supply-chain limits

Honors & Technical Skills

Honors & Awards: Samsung Semiconductor Research Fellowship ($10K), Samsung Semiconductor Intern Scholarship ($10K), Humba Deep Tech Fellowship ($6K), TEX-E Fellowship
Characterization: Scanning Electron Microscopy, X-Ray Diffraction, Thermogravimetric Analysis, Optical Microscopy, FTIR
Engineering Modeling: COMSOL Multiphysics, Aspen Plus, LOPA, Process Hazard Analysis, Fusion 360
Programming & Data: Python (NumPy, Pandas, SciPy, PyTorch, TensorFlow), SQL, MATLAB, Git, Docker, Linux